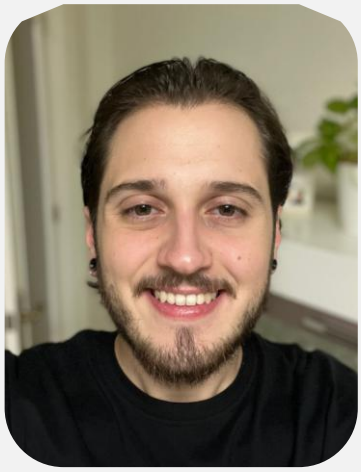


CONTACT

rafaroman182@gmail.com

+34 665 195 230

Seville, Spain

github.com/rafaroman18

linkedin.com/in/rafaroman18

GENERAL SKILLS

- Data Visualization
- Deep Learning
- Machine Learning
- Agile Methodologies
- Data Processing
- ETL Pipelines
- Generative Models (GANs)
- Relational & NoSQL Databases

TECHNICAL SKILLS

Languages: Python, Pandas, PyTorch, Scikit-Learn, Keras, TensorFlow, Matplotlib, C#, Java

Databases: Spark, Hadoop, MongoDB, Powercenter, QLIK, Oracle, Teradata, Microsoft SQL Server, SQL, NO-SQL

Cloud: AWS, GCP, Azure
Other: Git, Docker, ElasticSearch

LANGUAGES

Spanish – **Native**
English – **B2 (Cambridge)**

CERTIFICATIONS

B2 First (FCE) - Cambridge
UiPath RPA Associate

Rafael Román Aguilar

Data Scientist · Generative AI & Deep Learning

Data Scientist with experience building **Deep Learning**, **Machine Learning** and **generative AI** solutions in Python. Skilled at analyzing and optimizing data flows, and currently designing, validating and deploying AI models in backend environments at FIDESOL. Adaptable, professional and persistent.

EDUCATION

Master's in Data Engineering and Science

National Distance Education University (UNED)
10/2022 – 06/2026

Bachelor's in Computer Engineering – Computing

School of Engineering (Universidad de Cádiz) – **Average Grade: 7.1/10**
09/2016 – 03/2021

EXPERIENCE

Data Scientist – FIDESOL

12/2024 – Present — Granada, Spain

AI research and model design, including evaluation and validation. Defined AI solution **architectures** and deployed models in **backend environments** using Python and Django REST Framework. Delivered **generative AI** virtual patients for medical training (UNIR), **LSTM** demand forecasting and heat stress detection with wearables.
Technologies: Python, Django, Docker, TensorFlow, PyTorch, LLMs, Azure ML

PLSQL Developer – Minsait

08/2022 – 12/2024 — Sevilla, Spain

Maintained and developed the processes that feed the **databases** of a banking client. Detected and fixed incidents, analyzed **data flows** and built transformations over **large data volumes**.
Technologies: Powercenter, QLIK, OBIEE, SQL Server, Oracle, Teradata

Data Scientist – NASSAT

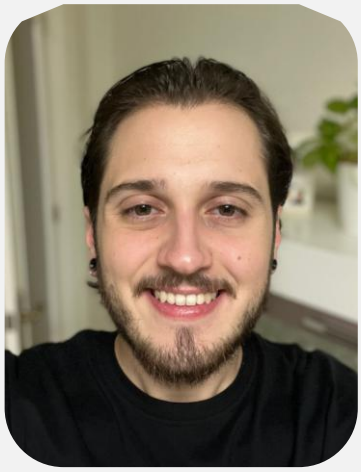
03/2022 – 04/2022 — Mijas, Spain

Built a **Deep Learning** model to predict wildfires, assign each zone a risk index and visualize it in a virtual reality headset. **Prepared and processed the data** with **Python, Pandas** and **NumPy** and built the model with **PyTorch**, working with geographic and historical data (.shp, .geoJSON).
Technologies: Python, PyTorch, Pandas, NumPy

DevOps Developer – Sopra Steria

04/2021 – 03/2022 — Sevilla, Spain

Worked for a major Andalusian client on large projects, **managing application databases**, deploying websites with **Docker** and monitoring scripts with **OpenShift**.
Technologies: Java, Python, Docker, ElasticSearch



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PROJECTS

Master's Thesis: GANs for Breast Cancer Detection
github.com/rafaroman18/BreastCancerDL

Studied whether a **StyleGAN** trained on only 260 thermal images can generate samples that improve a breast cancer classifier. Bias-aware design with **400 StyleGAN models**, 5 augmentation settings and **5-fold cross-validation** repeated 10 times. Adding about 50% synthetic images gave the best trade-off in recall.

Heat Stress Detection with a Smart Wristband (FIDESOL)

Prototype to detect heatstroke in workers in extreme environments. Trained a **Machine Learning** model (**decision trees**) on wristband data. When heat stress is detected, an alert is sent to a control center that logs every incident, and the wristband signals with sound and light that the worker must rest and notify a supervisor.

Bus Passenger Demand Forecasting (FIDESOL)

Predictive model of daily passenger volume so a bus company can plan extra buses in advance. Built an **LSTM** network and a **regression** model, the LSTM delivering the best results. Bus data and predictions are stored in a database and the client can **retrain** the models. Responsible for data analysis and cleaning and model design.

Airbnb Apartment Classification and Regression
github.com/rafaroman18/AIRBNB_ML

Personal project to study the **housing market**. Classified the room type of each apartment and predicted its price from an Airbnb dataset of the Community of Madrid, covering the full data analysis and processing workflow and comparing **Naive Bayes**, **KNN**, **decision trees** and **Support Vector Machines (SVM)**.

Virtual Training Environments Prototype (LeapVR)
github.com/rafaroman18/LeapVR

Virtual reality environment for corporate training, viewed with **Oculus Rift** and controlled without handheld controllers through **Leap Motion**. With **Alexa**, simulations could be launched and 3D models from an online collection added on the fly.

Telegram Bot on a Raspberry Pi (TorrentRaspBot)
github.com/rafaroman18/TorrentRaspBot

Bot built for personal use that downloads any link or torrent (magnet links or .torrent files) sent through **Telegram** with a command such as "/download", reports the progress and uploads the files to **Google Drive**. Several users showed interest in it.